

FAKE PRODUCT DETECTION SYSTEM
A MINI-PROJECT REPORT

Submitted by
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in partial fulfillment of the award of the degree
of
BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)



RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
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BONAFIDE CERTIFICATE

Certified that this project “**FAKE PRODUCT DETECTION SYSTEM**” is the bonafide work of “**M JANANY**” who carried out the project work under my supervision.

This mini project report is submitted for the viva voce examination to be held on _____

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ACKNOWLEDGEMENT

We express our sincere thanks to our beloved and honorable chairman **MR. S. MEGANATHAN** and the chairperson **DR. M.THANGAM MEGANATHAN** for their timely support and encouragement.

We are greatly indebted to our respected and honorable principal **Dr. S.N. MURUGESAN** for his able support and guidance.

No words of gratitude will suffice for the unquestioning support extended to us by our Head Of The Department **Mr.J N Benedict** for being ever supporting force during our project work

We also extend our sincere and hearty thanks to our internal guide **Mrs.Jananee**, for her valuable guidance and motivation during the completion of this project.

Our sincere thanks to our family members, friends and other staff members of computer science engineering.

1. M JANANY (231901012)

ABSTRACT

The **Fake Product Detection System** is an innovative solution designed to combat the growing problem of counterfeit goods in the market. By leveraging the decentralized and immutable nature of **Blockchain technology**, this system ensures that each product's authenticity can be verified by end-users in real time. Manufacturers generate unique **QR codes** for each product, which are then linked to product data securely stored on a blockchain network using **Ethereum smart contracts**.

When scanned by customers, the QR code retrieves the product details from the blockchain, allowing users to instantly verify whether the product is genuine. This enhances **transparency**, builds **consumer trust**, and significantly reduces the risk of fraud.

In addition to blockchain, the system incorporates **Operating System (OS) concepts** such as **CPU Scheduling Algorithms** (FCFS, SJF, Round Robin) and **Banker's Algorithm** to simulate efficient task handling and resource allocation in the backend process. These simulations not only optimize system performance but also demonstrate how real-world OS techniques can be integrated into web-based solutions.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In today's global marketplace, counterfeit products are a major concern, affecting industries ranging from pharmaceuticals and electronics to consumer goods. These fake products not only lead to economic losses but also pose serious risks to consumer health and safety. This system enables manufacturers to create a **unique QR code** for each genuine product, which is linked to its information stored on the blockchain via **smart contracts**. When a consumer scans the QR code, the system verifies the product's authenticity based on the stored blockchain data.

1.2 SCOPE OF THE WORK

The scope of this project includes:

- Developing a user-friendly interface for **Manufacturers** and **Customers**.
- Generating and scanning **QR codes** for product identification.
- Interacting with **Ethereum smart contracts** for storing and verifying product data.
- Simulating **Operating System algorithms** to demonstrate process scheduling and resource allocation concepts.
- Providing graphical representation of OS performance to analyze algorithm efficiency.

This project is designed to showcase a real-world application of cybersecurity and OS concepts while addressing the issue of product counterfeiting.

1.3 PROBLEM STATEMENT

With the rapid increase in counterfeit products, consumers often find it difficult to differentiate between genuine and fake items. Traditional methods of verification are prone to tampering and lack transparency. There is a pressing need for a **decentralized and secure system** that allows users to easily verify a product's authenticity and ensure its origin.

1.4 AIM AND OBJECTIVES OF THE PROJECT

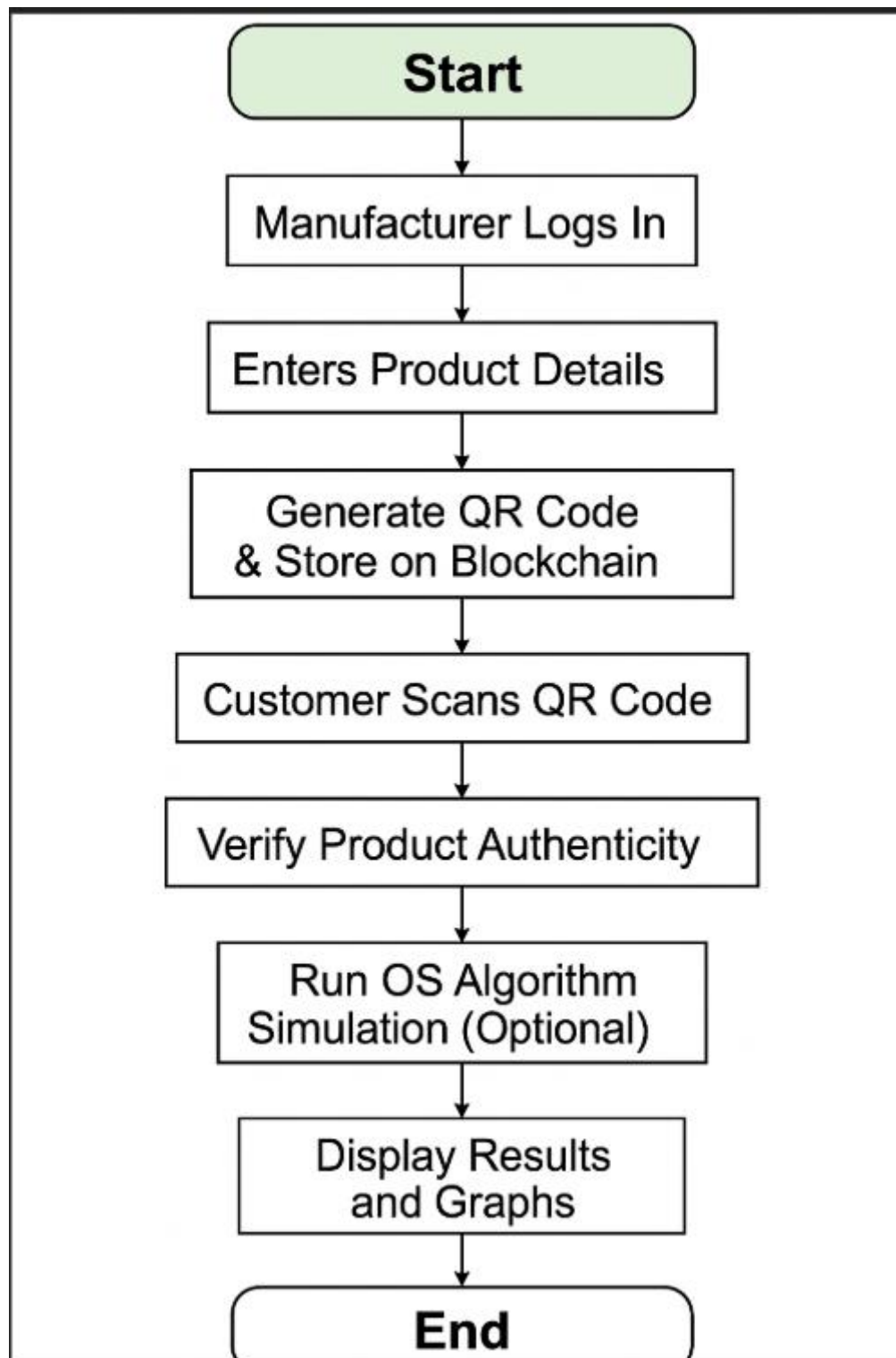
Aim:

To develop a decentralized system that enables the verification of product authenticity using Blockchain technology and enhances system performance visualization using Operating System algorithms.

Objectives:

- To generate a **QR code** for every genuine product containing product-specific data.
- To store product data on the **blockchain** using **smart contracts** for tamper-proof verification.
- To allow customers to scan QR codes and verify products instantly.
- To simulate and compare the performance of **FCFS, SJF, and Round Robin CPU scheduling algorithms**.
- To implement **Banker's Algorithm** to showcase safe resource allocation.
- To visually display performance metrics using graphs for better analysis.

FLOWCHART



CHAPTER 2

SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	:	Ryzen 5
Memory Size	:	8GB (Minimum)
HDD	:	1 TB (Minimum)

2.2 SOFTWARE SPECIFICATIONS

- **Operating System** : Windows 11
- **Front-End** : HTML, CSS, JavaScript
- **Back-End** : Smart Contracts (Solidity using Remix IDE)
- **Languages Used** : JavaScript, Solidity
- **Blockchain Tools** : Ethereum (Ganache), MetaMask, Ethers.js
- **OS Concepts Integrated** : CPU Scheduling (FCFS, SJF, Round Robin), Banker's Algorithm
- **Charting Library** : Chart.js
- **Other Tools** : QR Code Generator (qrcode.min.js), Visual Studio Code (for development)

CHAPTER 3

MODULE DESCRIPTION

The system is divided into the following functional modules:

1. **Manufacturer Module**

- Adds product information and generates a QR code.
- Deploys product data to the blockchain using smart contracts.

2. **Customer Module**

- Scans the QR code to retrieve product ID.
- Verifies the product on blockchain and shows authenticity.

3. **Operating System Module**

- Simulates CPU Scheduling (FCFS, SJF, Round Robin) and Banker's Algorithm.
- Visualizes performance with comparison graphs.

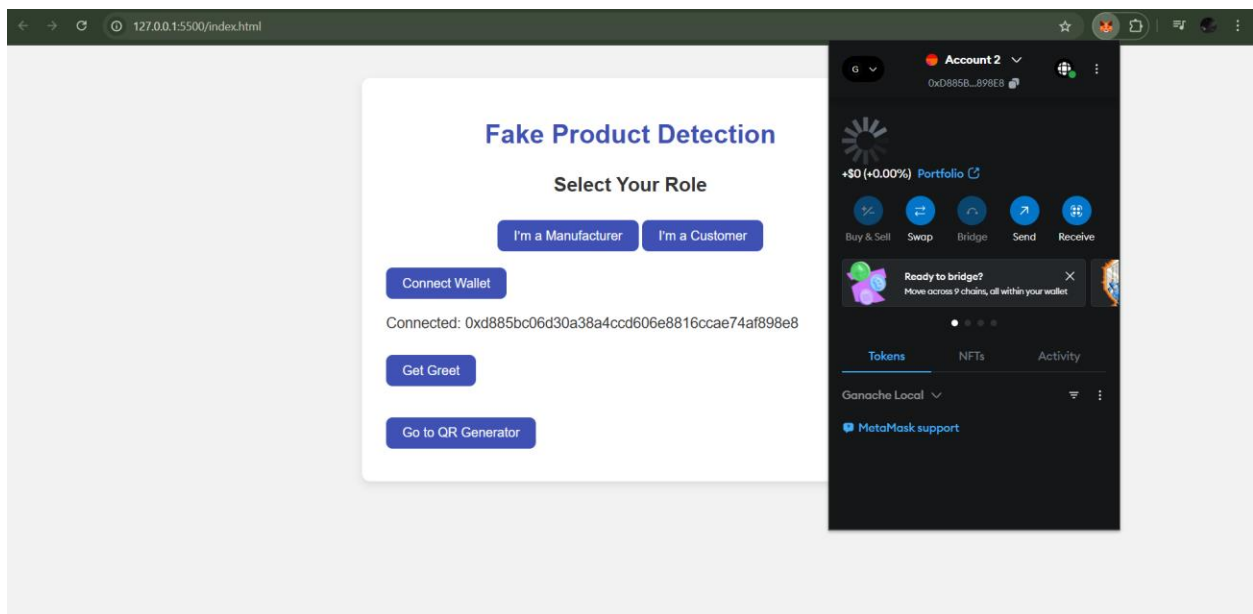
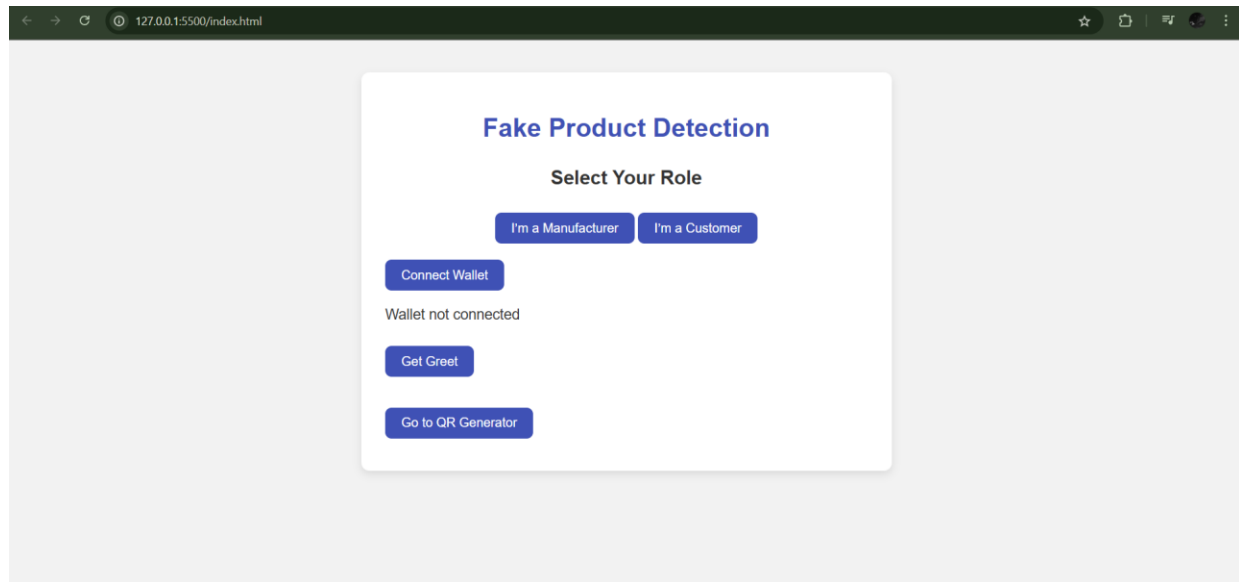
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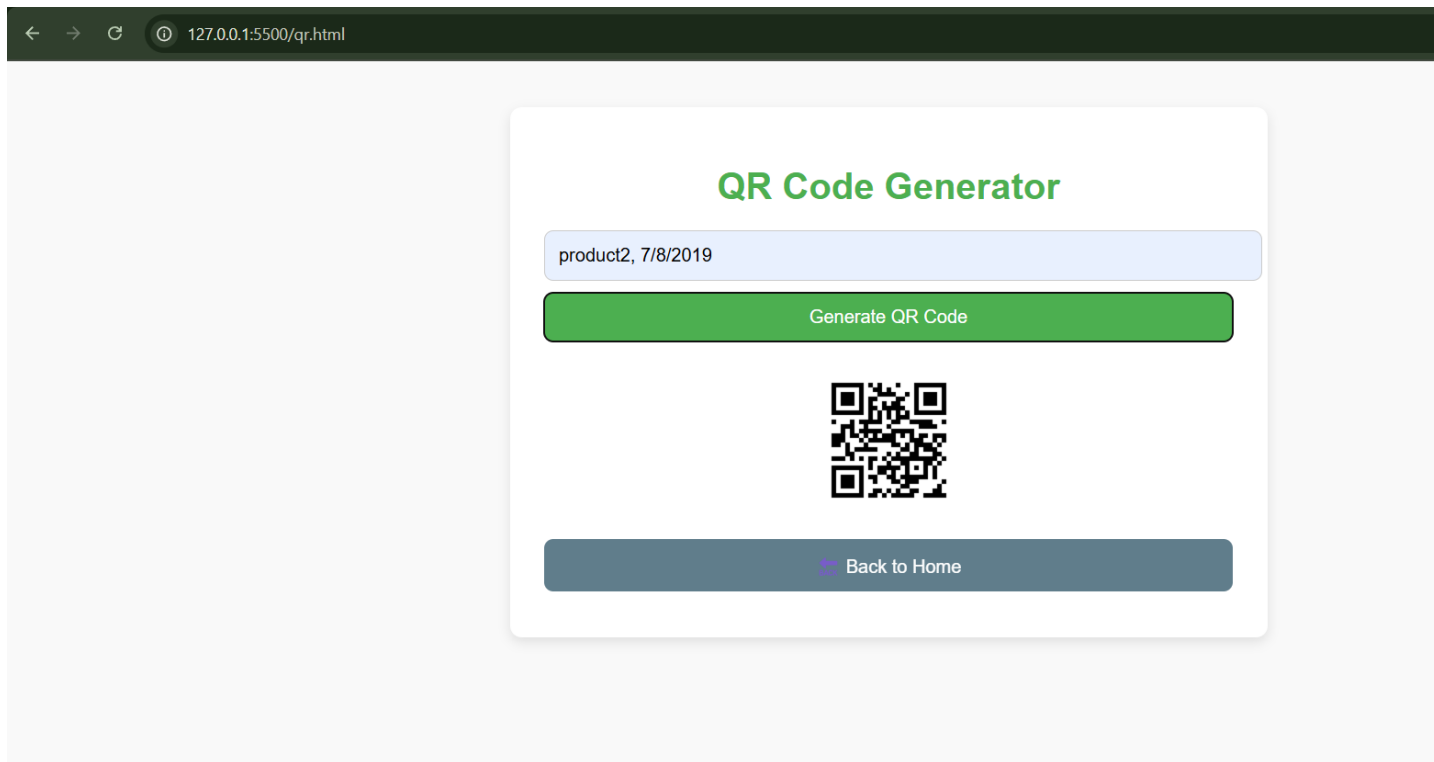
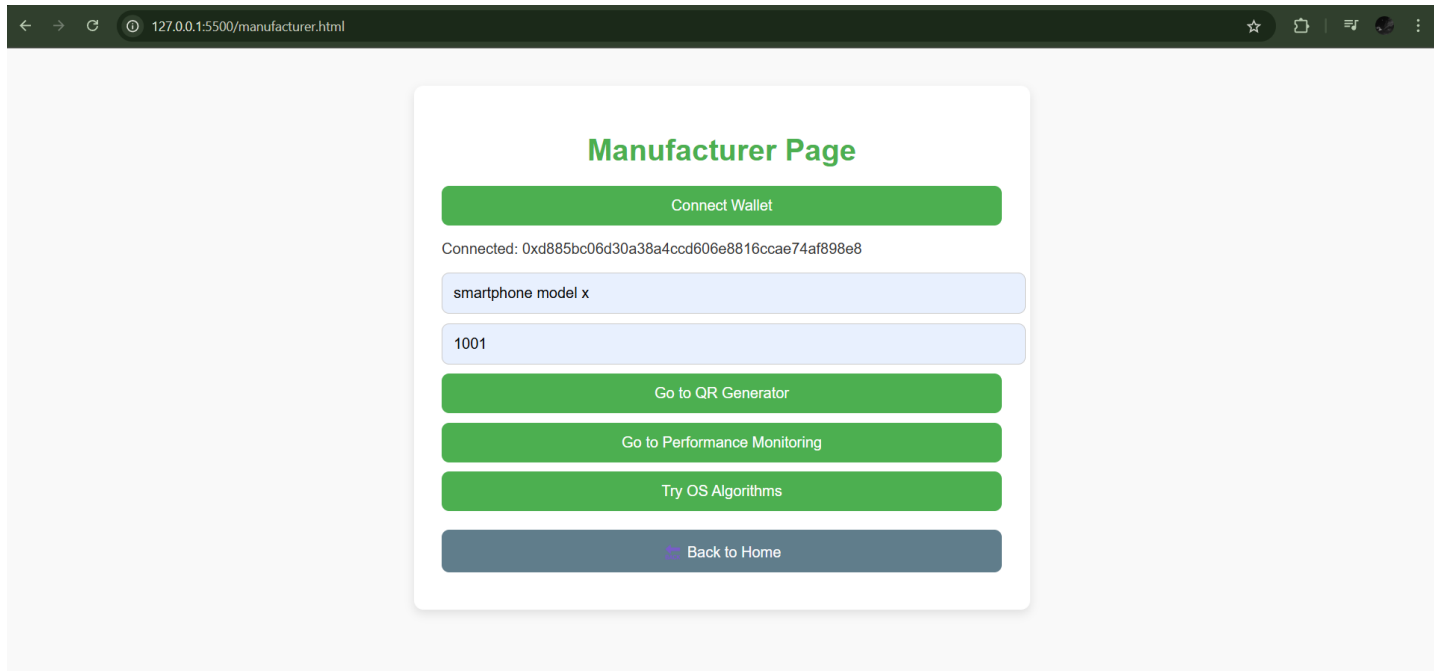
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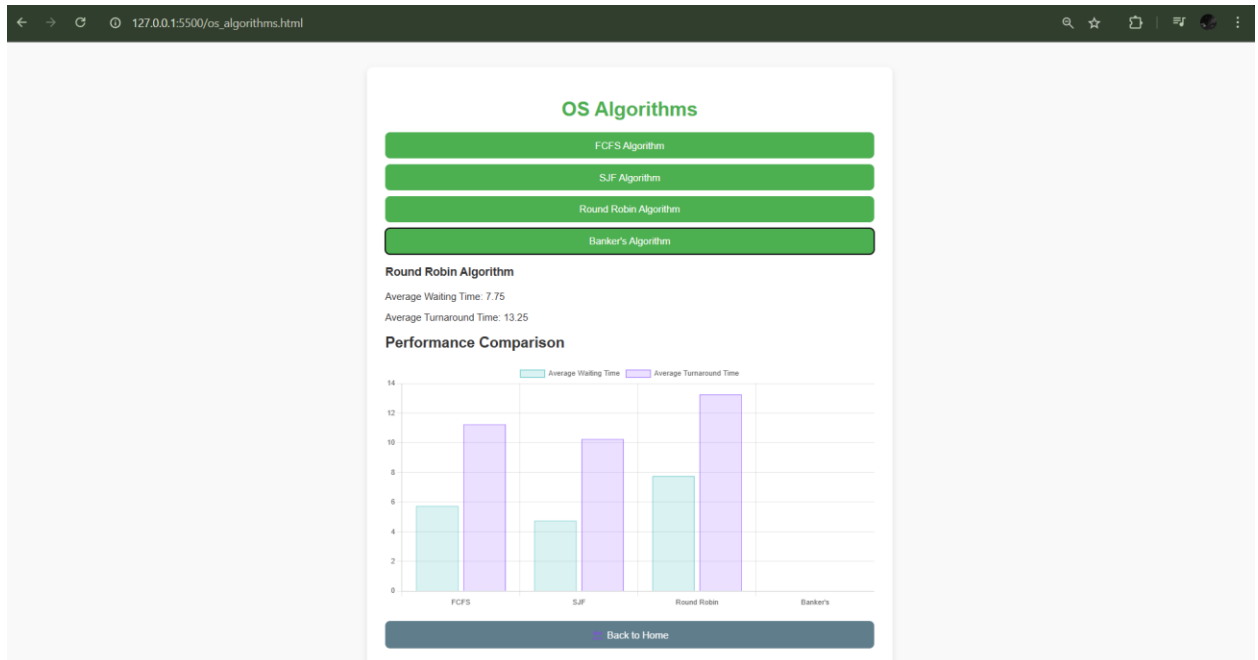
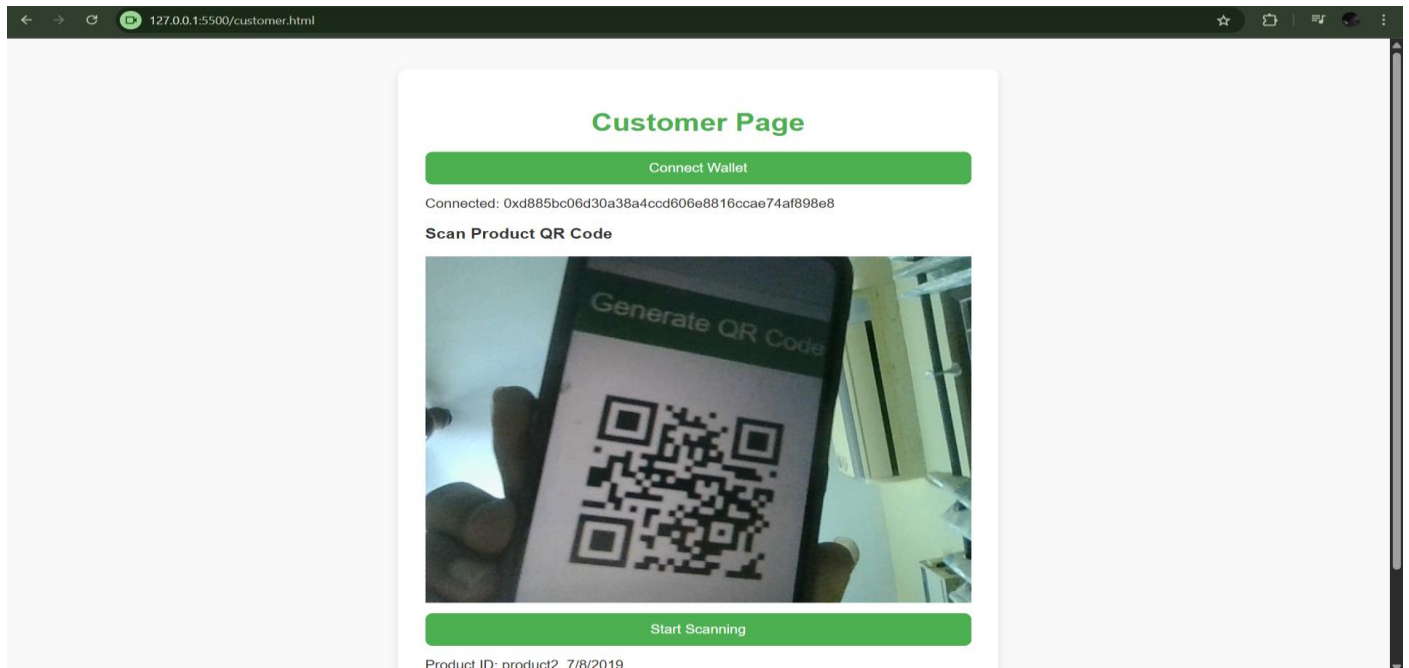
- **index.html**: Main user interface with options for Manufacturer and Customer. Handles navigation to role-specific pages.
- **manufacturer.html**: UI page for Manufacturer to enter product details and generate QR codes.
- **customer.html**: UI page for Customer to scan QR code and verify product authenticity via blockchain.
- **generate.js**: JavaScript logic for QR code generation and form handling in manufacturer page.
- **verify.js**: JavaScript logic for QR code scanning and product verification in customer page using Ethers.js and MetaMask.
- **smart_contract.sol**: Solidity smart contract deployed on Ethereum (via Remix & Ganache) to store and retrieve product details.
- **os_algo.js**: Implements CPU Scheduling Algorithms (FCFS, SJF, Round Robin) and Banker's Algorithm with chart visualization.
- **chart.js**: Used to render comparison graphs for scheduling algorithms' performance.
- **qrcode.min.js**: External library used to generate QR codes for products.
- **styles.css**: CSS file for styling all UI pages.
- **requirements.txt**: Lists necessary libraries such as Ethers.js (included via CDN), qrcode.min.js, and chart.js.

CHAPTER 5

SCREENSHOTS







CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion:

The Fake Product Detection System integrates blockchain for secure product validation and operating system concepts for educational simulation. The decentralized structure improves transparency and builds customer trust.

Future Enhancement:

- Upgrade the database to use IPFS for distributed file storage.
- Implement Admin dashboard to monitor system metrics.
- Integrate SMS/Email alerts for verified scans.
- Extend OS simulation with more algorithms like Priority Scheduling and Deadlock Detection.

REFERENCES

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2. **Remix IDE**
3. **MetaMask Docs**
4. **Chart.js Documentation**
5. **Operating System Concepts – Abraham Silberschatz**
6. **Blockchain Basics – Daniel Drescher**